Infectious Diseases I: UTI, Respiratory Infections, Tuberculosis, Brucellosis

All key points for the exam (IDSA, ATS/IDSA, CDC, AAP, WHO) — drug table after each topic

0. General Antimicrobial Principles (apply to every ID topic)

- **Get cultures before antibiotics** (without delaying therapy in sepsis/meningitis) • **empiric → definitive** (de-escalate to narrowest effective agent).
- Choose by: **site of infection**, likely organisms, **local resistance** (antibiogram), patient factors (allergy, renal/hepatic function, pregnancy, age, DDIs), drug factors (penetration, bioavailability, cost).
- **Concentration-dependent killing** (aminoglycosides, fluoroquinolones, daptomycin): high peaks, once daily • **time-dependent** (β-lactams): maximize time above MIC (frequent dosing, extended infusion) • **AUC-dependent** (vancomycin, FQs, azithromycin).
- **High oral bioavailability** (IV = PO): **fluoroquinolones**, **TMP-SMX**, **linezolid**, **metronidazole**, **doxycycline/minocycline**, **fluconazole**, voriconazole, rifampin, clindamycin.
- **Penicillin allergy:** ~90% of labeled patients are not truly allergic → allergy history/testing • cross-reactivity with cephalosporins is low (~1–2%, side-chain related); carbapenems < 1%; **aztreonam no cross-reactivity** (except ceftazidime — shared side chain).
- **Chelation:** fluoroquinolones and tetracyclines bind **Ca, Mg, Al, Fe, Zn** (antacids, dairy for tetracyclines, multivitamins, sucralfate) → separate (FQ 2 h before or 6 h after).

PART A — URINARY TRACT INFECTIONS (UTI)

1. Classification, Microbiology and Diagnosis

Type	Definition
Asymptomatic bacteriuria (ASB)	Significant bacteriuria **without symptoms**
Uncomplicated cystitis	Lower UTI (bladder) in **healthy, non-pregnant women** without urologic abnormalities
Complicated UTI	UTI with factors that ↑ failure risk: **men**, **pregnancy**, structural/functional abnormality, **catheter**, obstruction (stones), immunosuppression, renal impairment, recent instrumentation • **IDSA 2025: any UTI extending beyond the bladder** (pyelonephritis, bacteremia, systemic signs)
Pyelonephritis	Upper UTI (kidney): **fever**, **flank pain**, **costovertebral angle (CVA) tenderness**, N/V ± cystitis symptoms
Recurrent UTI	**≥ 2 infections in 6 months** or **≥ 3 in 12 months** (relapse = same organism within 2 weeks; reinfection = new organism)
CAUTI	Catheter-associated UTI (catheter in place > 2 days or removed within 48 h)

Microbiology

- **Escherichia coli** (75–95% of uncomplicated) > **Staphylococcus saprophyticus** (young sexually active women) > **Klebsiella**, **Proteus mirabilis** (urease → **struvite stones**, alkaline urine; **intrinsically resistant to nitrofurantoin**), Enterococcus.
- **Complicated/healthcare-associated:** resistant gram-negatives (**ESBL** E. coli/Klebsiella, **Pseudomonas**, Enterobacter, Serratia), **enterococci**, Candida.
- **S. aureus in urine** → think **bacteremia** (hematogenous seeding) → blood cultures.

Symptoms and diagnosis

- **Cystitis: dysuria**, **frequency**, **urgency**, suprapubic pain, hematuria • no vaginal discharge (if present consider vaginitis/STI).

- **Elderly:** atypical (confusion, falls) — but **confusion alone with bacteriuria is not an indication to treat** without systemic signs.
- **Urinalysis: pyuria** (≥ 10 WBC/mm³), **leukocyte esterase** (+), **nitrite** (+) — gram-negatives convert nitrate → nitrite (**Enterococcus, S. saprophyticus, Pseudomonas are nitrite-negative**), hematuria, bacteria, **WBC casts = pyelonephritis**.
- **Urine culture:** not needed for uncomplicated cystitis in women • **required** for pyelonephritis, complicated UTI, men, pregnancy, recurrent, treatment failure • significant: **≥ 10^5 CFU/mL** (≥ 10^3 in symptomatic women/men or catheterized).
- **Clean-catch midstream** specimen • collect before antibiotics • blood cultures in pyelonephritis with sepsis.

2. Treatment of UTI

Acute uncomplicated cystitis (IDSA)

Drug	Regimen	Notes
Nitrofurantoin monohydrate/macrocrystals (Macrobid)	**100 mg PO BID × 5 days**	**First-line** • with food • no systemic levels → **NOT for pyelonephritis** • avoid CrCl < 30 (Beers) • brown urine
TMP-SMX DS (160/800 mg)	**1 tab BID × 3 days**	First-line if local resistance **< 20%** and no use in prior 3 months
Fosfomycin	**3 g sachet PO once** (dissolve in water)	First-line • slightly less effective • diarrhea • not for pyelonephritis
Pivmecillinam	185 mg PO TID × 3–7 days	First-line in guidelines (FDA-approved 2024) • penicillin (avoid in PCN allergy)
Gepotidacin	1,500 mg PO BID × 5 days	New (2025) triazaacenaphthylene • GI effects, QT caution • reserve
β-lactams (amoxicillin-clavulanate, cefpodoxime, cefdinir, cephalexin)	**5–7 days**	**Second-line** (less effective, collateral damage) • amoxicillin/ampicillin alone not recommended (resistance)
Fluoroquinolones (ciprofloxacin 250 mg BID, levofloxacin 250 mg daily)	**3 days**	**Reserve** — not first-line for cystitis (FDA boxed warnings, resistance) • save for pyelonephritis

- **Phenazopyridine** 200 mg TID × **2 days** (urinary analgesic) — symptomatic relief only • **orange-red urine**, stains contact lenses/clothes, hemolysis in **G6PD** deficiency, methemoglobinemia, avoid CrCl < 50.
- Increase fluids • symptoms should improve in 48–72 h; if not → culture and reassess.

Acute pyelonephritis

- **Outpatient (mild–moderate, can take PO): ciprofloxacin 500 mg BID × 7 days** (or 1,000 mg ER daily) or **levofloxacin 750 mg daily × 5–7 days** if local FQ resistance ≤ 10% • if resistance > 10% → give an initial **IV dose of ceftriaxone 1 g** (or a consolidated 24-h aminoglycoside dose) then FQ.
- **TMP-SMX DS BID × 14 days** (7–10 days in newer guidance) if organism susceptible (initial ceftriaxone dose if empiric) • **oral β-lactams** less effective → initial IV ceftriaxone + 10–14 days.
- **Inpatient: ceftriaxone 1–2 g IV daily**, cefepime, **piperacillin-tazobactam**, FQ IV, aminoglycoside ± ampicillin • **ESBL risk → carbapenem** (ertapenem 1 g daily, meropenem) • **sepsis/shock → broad (carbapenem or newer BL/BLI if MDR risk)** • switch to oral when improving (afebrile 48 h).
- **Nitrofurantoin and fosfomycin should NOT be used for pyelonephritis** (inadequate renal tissue levels).
- **IDSA 2025 complicated UTI: 7 days** total is enough for most patients with prompt improvement (vs 10–14) • early IV-to-oral switch with highly bioavailable agents.

Special situations

Situation	Management
Asymptomatic bacteriuria (IDSA 2019)	**Do NOT screen or treat** — except **pregnancy** (screen at 12–16 weeks, treat 4–7 days) and **before urologic procedures with mucosal trauma** (TURP) • do not treat in elderly, diabetes, catheters, spinal cord injury, renal transplant > 1 month, pyuria alone
Pregnancy (ASB or cystitis)	**Nitrofurantoin** (avoid 1st trimester if alternatives and at term — hemolytic anemia in newborn), **cephalexin**, **amoxicillin-clavulanate**, cefpodoxime, **fosfomycin** • 5–7 days • **avoid TMP-SMX in 1st trimester (folate antagonist) and near term (kernicterus)** • **avoid fluoroquinolones and tetracyclines** • test of cure culture
Pyelonephritis in pregnancy	**Hospitalize** → IV **ceftriaxone** (or cefepime, pip-tazo) → oral to complete 10–14 days → **suppressive therapy** (nitrofurantoin or cephalexin nightly) for rest of pregnancy
Men	All UTIs in men are complicated • cystitis **7 days** (TMP-SMX, FQ; nitrofurantoin acceptable for cystitis only) • check prostate
Acute bacterial prostatitis	**TMP-SMX or FQ** (good prostate penetration) **2–4 weeks** • severe → IV ceftriaxone/FQ then oral
Chronic bacterial prostatitis	**FQ 4–6 weeks** (preferred) or TMP-SMX 1–3 months
CAUTI	**Remove or replace the catheter** (if in place ≥ 2 weeks) and culture from the new catheter • **7 days** if prompt response, **10–14 days** if delayed • levofloxacin 5 days if not severe • 3 days in women ≤ 65 after catheter removal • prevention: avoid unnecessary catheters, remove ASAP
Recurrent UTI prevention	Behavioral (hydration, post-coital voiding — limited evidence) • **vaginal estrogen in postmenopausal women** • **continuous prophylaxis** 6–12 months: **nitrofurantoin 50–100 mg nightly**, **TMP-SMX SS (½ DS) nightly or 3×/week**, cephalexin 125–250 mg, fosfomycin 3 g every 10 days • **post-coital single dose** • **self-start therapy** • **methenamine hippurate 1 g BID** (needs acidic urine, avoid with sulfonamides) • cranberry, D-mannose (modest) • avoid spermicides (diaphragm)
Candiduria	Usually colonization (catheter) → remove catheter; treat only if symptomatic, neutropenic, or before urologic procedure (fluconazole)
Children	Febrile infants → culture by catheter • oral cephalosporins (cefixime, cephalexin), amoxicillin-clavulanate, TMP-SMX • 7–14 days • avoid FQs (cartilage) • imaging (renal ultrasound) after first febrile UTI

3. UTI Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Nitrofurantoin (Macrobid)	**100 mg BID × 5 d** • prophylaxis 50–100 mg nightly	GI, **pulmonary toxicity** (acute/chronic fibrosis), **hepatotoxicity**, **peripheral neuropathy**, hemolysis (G6PD)	**Not for pyelonephritis** • avoid CrCl < 30 • with food • brown urine • Proteus resistant
TMP-SMX	**DS BID × 3 d** (cystitis) • 14 d (pyelo)	**Hyperkalemia**, rash/**SJS**, ↑ SCr (blocks secretion), bone marrow suppression, photosensitivity	**↑ INR with warfarin** (2C9) • ↑ K with ACEi/ARB/spironolactone • ↑ methotrexate toxicity • sulfa allergy • avoid 1st trimester/term
Fosfomycin	**3 g once**	Diarrhea, headache	Not for pyelonephritis • ESBL/VRE cystitis option
Ciprofloxacin	Cystitis 250 mg BID × 3 d • pyelo **500 mg BID × 7 d**	**Boxed: tendinitis/rupture, neuropathy, CNS, MG** • QT, dysglycemia, aortic aneurysm, C. diff	**Chelation** (Ca, Mg, Al, Fe, Zn) • **CYP1A2 inhibitor** (↑ theophylline, tizanidine — CI) • avoid pregnancy/children
Levofloxacin	Cystitis 250 mg × 3 d • pyelo **750 mg daily × 5–7 d**	Same	Renal adjustment • chelation • ↑ INR

Drug	Dose	Key ADRs	Key notes / interactions
Cephalexin	250–500 mg q6–12h × 5–7 d	GI, rash	Pregnancy-safe
Amox-clavulanate	500/125 mg BID–TID or 875/125 BID × 5–7 d	**Diarrhea**, rash	Pregnancy-safe
Cefpodoxime	100 mg BID × 5–7 d (pyelo 200 mg BID × 10–14 d)	GI	Absorption ↓ with antacids/PPIs
Ceftriaxone	1–2 g IV daily	Biliary sludge, C. diff	Initial dose before oral therapy • **no Ca-IV fluids in neonates**
Ertapenem	1 g IV/IM daily	Seizures, C. diff	**ESBL** • no Pseudomonas/Acinetobacter/Enterococcus • ↓ **valproate**
Pivmecillinam	185 mg TID × 3–7 d	GI, carnitine depletion (long term)	Penicillin class
Phenazopyridine	200 mg TID × 2 d	**Orange urine**, hemolysis (G6PD), methemoglobinemia	Symptom relief only • stains contacts
Methenamine hippurate	1 g BID	GI, rash	Prophylaxis • needs acidic urine • avoid with sulfonamides, CrCl < 50

PART B — RESPIRATORY TRACT INFECTIONS

4. Upper Respiratory Tract Infections

Acute pharyngitis

- **Mostly viral** (cough, rhinorrhea, hoarseness, conjunctivitis, oral ulcers suggest viral) • bacterial = **Group A Streptococcus (GAS, S. pyogenes)** (5–15% adults, 20–30% children 5–15 years).
- **Centor/McIsaac criteria:** fever > 38 °C, **tonsillar exudates/swelling**, **tender anterior cervical lymphadenopathy**, **absence of cough** (+1 age 3–14, −1 age ≥ 45) • score ≥ 2–3 → test.
- **Diagnosis: rapid antigen detection test (RADT)** or NAAT • **negative RADT in children/adolescents → confirm with throat culture** (not needed in adults).
- **Goals of antibiotics:** prevent **acute rheumatic fever** (can start within 9 days), ↓ suppurative complications (peritonsillar abscess), ↓ duration, ↓ transmission.
- **Treatment duration: 10 days** (except azithromycin 5 days and benzathine single dose) • **no penicillin resistance in GAS**.
- Symptomatic: acetaminophen/ibuprofen, lozenges • **avoid aspirin in children** (Reye's) • contagious until 12–24 h of antibiotics.

Patient	Regimen
No allergy (first-line)	**Penicillin V** 250 mg BID–QID (children) / **500 mg BID** (adults) × 10 days • **amoxicillin 50 mg/kg once daily (max 1,000 mg)** × 10 days • **benzathine penicillin G 1.2 million units IM once** (600,000 units if < 27 kg) — adherence concerns
Penicillin allergy — non-anaphylactic	**Cephalexin** 20 mg/kg BID (max 500 mg) × 10 days, **cefadroxil** 30 mg/kg daily (max 1 g) × 10 days
Penicillin allergy — anaphylactic	**Clindamycin** 7 mg/kg TID (max 300 mg) × 10 days • **azithromycin** 12 mg/kg daily (max 500 mg) × **5 days** (adults 500 mg day 1 then 250 mg days 2–5) • clarithromycin 10 days (macrolide resistance possible)

Acute otitis media (AOM — AAP)

- Pathogens: **S. pneumoniae**, **non-typeable H. influenzae** (β-lactamase), **Moraxella catarrhalis** (β-lactamase) • diagnosis: **bulging tympanic membrane**, otalgia, fever, otorrhea.

- **Observation option** (48–72 h, if follow-up assured): age 6–23 months with non-severe **unilateral** AOM • ≥ 24 months non-severe unilateral or bilateral.
- **Antibiotics immediately:** < 6 months • severe symptoms (temp ≥ 39 °C, moderate-severe otalgia, > 48 h) • otorrhea • bilateral AOM at 6–23 months.
- **First-line: high-dose amoxicillin 80–90 mg/kg/day divided BID** (overcomes intermediate-resistant pneumococcus).
- **Amoxicillin-clavulanate 90 mg/kg/day** (of amoxicillin, ES-600 formulation, divided BID) if **amoxicillin in past 30 days**, concurrent **purulent conjunctivitis** (H. influenzae), or recurrent AOM unresponsive to amoxicillin.
- **Penicillin allergy (non-severe): cefdinir** 14 mg/kg/day, **cefuroxime** 30 mg/kg/day, **cefpodoxime** 10 mg/kg/day, **ceftriaxone 50 mg/kg IM/IV × 1–3 days** (vomiting/failure) • severe allergy: clindamycin (no H. influenzae coverage), macrolides (poor).
- **Duration: 10 days if < 2 years** (or severe) • **7 days at 2–5 years • 5–7 days ≥ 6 years**.
- Treatment failure at 48–72 h → amoxicillin-clavulanate → ceftriaxone 3 days → clindamycin ± cephalosporin • pain: acetaminophen/ibuprofen • prevention: PCV, influenza vaccine, breastfeeding, avoid smoke, no pacifier.

Acute bacterial rhinosinusitis (IDSA)

- > 90–98% viral • suspect **bacterial** if: (1) **persistent symptoms ≥ 10 days** without improvement, (2) **severe** onset (fever ≥ 39 °C + purulent discharge/facial pain for ≥ 3–4 consecutive days), or (3) **"double sickening"** (worsening after initial improvement of a viral URI).
- Same pathogens as AOM • **first-line: amoxicillin-clavulanate** 500/125 mg TID or 875/125 mg BID (adults **5–7 days**; children 10–14 days) • **high-dose 2,000/125 mg BID** if risk of resistance (age ≥ 65, recent hospitalization or antibiotics within 1 month, immunocompromised, daycare, high local resistance).
- **Penicillin allergy: doxycycline** 100 mg BID or 200 mg daily (adults) or **respiratory FQ** (levofloxacin 500 mg, moxifloxacin 400 mg) • children: clindamycin + cefixime/cefpodoxime, levofloxacin.
- **Not recommended: macrolides and TMP-SMX** (high pneumococcal resistance) • **adjuncts: intranasal corticosteroids**, **saline irrigation** • avoid antihistamines/decongestants routinely.
- Watchful waiting acceptable in uncomplicated cases with follow-up.

Other upper respiratory infections

- **Common cold:** viral (rhinovirus) → **no antibiotics** (colored mucus is not bacterial) • symptomatic: decongestants, antihistamines (1st gen), NSAIDs, zinc (modest) • avoid OTC cough/cold products in children < 4 years.
- **Acute bronchitis:** mostly viral, cough up to 3 weeks → **no antibiotics** • **pertussis** (paroxysmal cough, whoop, post-tussive vomiting) → **azithromycin** (500 mg day 1, 250 mg days 2–5; preferred in infants < 1 month), clarithromycin, TMP-SMX (> 2 months) • treat close contacts.
- **Epiglottitis** (H. influenzae b) → ceftriaxone + vancomycin, airway.

Influenza

- **Treat as soon as possible (ideally < 48 h of symptoms)** — but **any time** for **hospitalized**, **severe/progressive** illness, or **high-risk** outpatients (age ≥ 65 or < 2, pregnancy and 2 weeks postpartum, chronic disease, immunosuppression, BMI ≥ 40, long-term care residents, < 19 on aspirin).
- **Oseltamivir** (neuraminidase inhibitor): **75 mg PO BID × 5 days • CrCl 30–60: 30 mg BID**, 10–30: 30 mg daily • preferred in **pregnancy** and hospitalized • ADRs: **nausea/vomiting** (take with food), neuropsychiatric events (children) • **prophylaxis 75 mg daily × 7 days** after exposure (10 days household).
- **Zanamivir** inhaled 10 mg BID × 5 days (age ≥ 7) • **bronchospasm → avoid in asthma/COPD**.
- **Peramivir** 600 mg IV single dose (outpatient unable to take PO).
- **Baloxavir marboxil** (cap-dependent endonuclease inhibitor): **single dose** 40 mg (40–79 kg) or 80 mg (≥ 80 kg), age ≥ 5 • **avoid with dairy, Ca, antacids, polyvalent cations** • not for pregnancy/hospitalized (limited data).
- **Amantadine/rimantadine not recommended** (resistance).
- **LAIV interaction:** antivirals given 48 h before to 2 weeks after LAIV reduce vaccine efficacy.

COVID-19 (outpatient, high risk of progression)

- **Nirmatrelvir/ritonavir (Paxlovid)**: **300/100 mg BID × 5 days** within 5 days of symptoms • **eGFR 30–60: 150/100 mg BID** • avoid eGFR < 30 (special dosing in dialysis per updated label) • **ritonavir = strong CYP3A4 inhibitor → major DDIs** (statins, DOACs, tacrolimus, amiodarone, antiarrhythmics, sildenafil, etc.) • ADRs: dysgeusia, diarrhea, rebound.
- **Remdesivir** IV 200 mg day 1 then 100 mg × 2 days (outpatient) • **molnupiravir** 800 mg BID × 5 days (alternative; not in pregnancy).
- Hospitalized with hypoxemia: **dexamethasone 6 mg daily × up to 10 days** ± remdesivir ± baricitinib/tocilizumab.

5. Community-Acquired Pneumonia (CAP — ATS/IDSA 2019)

- **Pathogens: Streptococcus pneumoniae** (most common), **H. influenzae**, **Moraxella**, **atypicals** (**Mycoplasma**, **Chlamydophila**, **Legionella**), respiratory viruses (influenza, RSV, SARS-CoV-2) • **S. aureus** (post-influenza, MRSA risk), **Pseudomonas** (structural lung disease, prior isolation).
- **Symptoms:** cough, fever, chills, **dyspnea**, pleuritic chest pain, sputum • crackles, ↓ breath sounds, egophony • **CXR infiltrate** required for diagnosis.
- **Site of care: PSI (Pneumonia Severity Index)** preferred over CURB-65 • **CURB-65**: **C**onfusion, **U**rea (BUN) > 19 mg/dL (7 mmol/L), **R**R ≥ 30, **B**P (SBP < 90 or DBP ≤ 60), age ≥ **65** → **0–1 outpatient**, **2 admit**, **≥ 3 consider ICU** (CRB-65 without BUN).
- **Severe CAP (ICU):** 1 major (**septic shock requiring vasopressors** or **respiratory failure requiring mechanical ventilation**) or ≥ 3 minor criteria (RR ≥ 30, PaO2/FiO2 ≤ 250, multilobar, confusion, BUN ≥ 20, WBC < 4,000, platelets < 100,000, T < 36 °C, hypotension needing aggressive fluids).
- **Diagnostics:** sputum and blood cultures in **severe CAP** or if empiric MRSA/Pseudomonas coverage/prior isolation/recent hospitalization with IV antibiotics • **urinary antigens** (pneumococcal, Legionella) in severe CAP • influenza/COVID testing • **procalcitonin should not be used to decide whether to start** antibiotics.

Empiric treatment of CAP

Setting	Recommended regimen
Outpatient — healthy, no comorbidities	**Amoxicillin 1 g TID or doxycycline 100 mg BID or** macrolide (**azithromycin 500 mg day 1 then 250 mg × 4 days**, or clarithromycin 500 mg BID / ER 1,000 mg daily) **only if local pneumococcal macrolide resistance < 25%**
Outpatient — with comorbidities (chronic heart, lung, liver, renal disease, diabetes, alcoholism, malignancy, asplenia)	**Combination: amoxicillin-clavulanate** (500/125 TID, 875/125 BID, or 2,000/125 BID) **or cephalosporin** (cefpodoxime 200 mg BID, cefuroxime 500 mg BID) **PLUS a macrolide or doxycycline • OR monotherapy with a respiratory fluoroquinolone** (**levofloxacin 750 mg daily**, **moxifloxacin 400 mg daily**, gemifloxacin 320 mg)
Inpatient — non-severe	**β-lactam** (**ampicillin-sulbactam 1.5–3 g q6h**, **ceftriaxone 1–2 g daily**, cefotaxime 1–2 g q8h, ceftaroline 600 mg q12h) **+ macrolide** (azithromycin 500 mg daily, clarithromycin) **OR respiratory FQ monotherapy** • doxycycline if macrolide and FQ contraindicated
Inpatient — severe (ICU)	**β-lactam + macrolide OR β-lactam + respiratory FQ** (no monotherapy)
Add MRSA coverage	Prior respiratory MRSA isolation, or recent hospitalization + IV antibiotics (90 days) with local risk • **vancomycin** (AUC 400–600) or **linezolid 600 mg q12h** • obtain cultures/nasal PCR → stop if negative (MRSA nasal PCR has high negative predictive value)
Add Pseudomonas coverage	Prior respiratory Pseudomonas, or recent hospitalization + IV antibiotics with local risk • **piperacillin-tazobactam 4.5 g q6h**, **cefepime 2 g q8h**, ceftazidime, **meropenem 1 g q8h**, imipenem, aztreonam

- **Duration: minimum 5 days** and until clinically stable (afebrile 48–72 h, stable vitals, eating, normal mentation) • **7 days** for MRSA/Pseudomonas • switch IV → PO when stable.
- **Aspiration pneumonia: do not routinely add anaerobic coverage** unless **lung abscess or empyema**.
- **Influenza-positive CAP:** add **oseltamivir** + standard antibacterials.

- **Corticosteroids:** not routine in non-severe CAP • **hydrocortisone (200 mg/day) may be used in severe CAP** (newer evidence) and in refractory septic shock.
- **Atypical clues: Legionella** (hyponatremia, diarrhea, high fever, ↑ LFTs, water sources) • **Mycoplasma** (young adults, cold agglutinins, bullous myringitis) → β-lactams do NOT cover atypicals (no cell wall) → need **macrolide, doxycycline, or FQ**.
- **Children CAP: amoxicillin 90 mg/kg/day** (outpatient); ampicillin/ceftriaxone inpatient; azithromycin if atypical suspected (> 5 years).
- **Prevention:** pneumococcal and influenza vaccines, smoking cessation.

6. Hospital-Acquired and Ventilator-Associated Pneumonia (IDSA 2016)

- **HAP** = pneumonia **≥ 48 h after admission** (not incubating) • **VAP** = pneumonia **≥ 48 h after endotracheal intubation** • (HCAP concept abandoned).
- **Pathogens: S. aureus (MRSA)**, **Pseudomonas aeruginosa**, Klebsiella, E. coli, Enterobacter, **Acinetobacter**, Serratia, Stenotrophomonas.
- **Use local antibiogram** • all empiric regimens cover **S. aureus and Pseudomonas**.

Coverage	When	Agents
MRSA	Prior IV antibiotics within 90 days, unit MRSA prevalence > 10–20% or unknown, high mortality risk (ventilation for HAP, septic shock)	**Vancomycin** 15 mg/kg q8–12h (load 25–30 mg/kg; AUC 400–600) **or linezolid 600 mg q12h** • otherwise MSSA coverage (pip-tazo, cefepime, levofloxacin, imipenem, meropenem)
Pseudomonas — one agent	All patients	**Piperacillin-tazobactam 4.5 g q6h** • **cefepime 2 g q8h** • ceftazidime 2 g q8h • **imipenem 500 mg q6h** • **meropenem 1 g q8h** • aztreonam 2 g q8h • levofloxacin 750 mg daily / ciprofloxacin 400 mg q8h
Pseudomonas — two agents (different classes)	IV antibiotics within 90 days, septic shock, ARDS, ≥ 5 days hospitalization before VAP, acute RRT, structural lung disease, high unit resistance (> 10%)	β-lactam above **+ aminoglycoside** (amikacin 15–20 mg/kg, gentamicin/tobramycin 5–7 mg/kg daily) **or FQ** or polymyxin • de-escalate to one agent once susceptibilities known

- **Duration: 7 days** for HAP/VAP (regardless of organism, if improving) • de-escalate based on cultures • procalcitonin + clinical criteria to stop.
- **Inhaled antibiotics** (aminoglycosides, colistin) as adjunct for VAP caused by gram-negatives susceptible only to aminoglycosides/polymyxins.
- **Daptomycin is inactivated by pulmonary surfactant → never for pneumonia** • **tigecycline** not recommended (↑ mortality).
- **Prevention of VAP:** head of bed 30–45°, daily sedation interruption/SBT, oral care, subglottic suctioning, minimize ventilator days.

7. Respiratory Infections Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Penicillin V	250–500 mg BID–QID × 10 d	Rash, GI	GAS pharyngitis (no resistance)
Benzathine penicillin G	**1.2 MU IM once** (600,000 if < 27 kg)	Injection pain, **never IV** (cardiac arrest)	GAS • syphilis
Amoxicillin	Pharyngitis 50 mg/kg (max 1 g) daily • **AOM 80–90 mg/kg/day** • CAP **1 g TID**	Rash (EBV mononucleosis), diarrhea	First-line AOM, CAP (healthy)
Amox-clavulanate	875/125 BID or 500/125 TID • high-dose 2,000/125 BID • AOM 90 mg/kg/day	**Diarrhea**, hepatotoxicity (cholestatic)	Sinusitis first-line • β-lactamase producers
Cephalexin	Pharyngitis 20 mg/kg BID (max 500)	GI	Non-anaphylactic PCN allergy

Drug	Dose	Key ADRs	Key notes / interactions
Cefdinir / cefpodoxime / cefuroxime	AOM 14 / 10 / 30 mg/kg/day • CAP cefpodoxime 200 BID	GI • cefdinir red stools with iron	Cefdinir/cefpodoxime ↓ with antacids, iron
Ceftriaxone	CAP 1–2 g daily • AOM 50 mg/kg IM × 1–3 d	Biliary sludge, C. diff	Neonates: avoid with Ca IV; hyperbilirubinemia
Azithromycin	**500 mg day 1 → 250 mg × 4 d** • inpatient 500 mg daily	GI, **QT prolongation**, hepatotoxicity	Atypicals • pertussis • fewer DDIs than clarithromycin
Clarithromycin	500 mg BID	Dysgeusia, GI, QT	**Strong CYP3A4 inhibitor** (statins, colchicine, DOACs, CCBs)
Doxycycline	100 mg BID	Photosensitivity, **esophagitis** (upright, water), tooth discoloration	**Chelation** • avoid pregnancy and < 8 years (short courses OK)
Levofloxacin	**750 mg daily** (CAP)	**Boxed: tendons, neuropathy, CNS, MG** • QT, dysglycemia	Respiratory FQ • renal adjust • chelation
Moxifloxacin	400 mg daily	Same + hepatotoxicity	**No renal adjustment • not for UTI** (low urine levels) • anaerobic activity
Amp-sulbactam	1.5–3 g IV q6h	Rash, diarrhea	CAP inpatient • aspiration/anaerobes
Ceftaroline	600 mg IV q12h	Rash, Coombs +	**Only cephalosporin with MRSA activity**
Pip-tazobactam	4.5 g IV q6h (or extended infusion)	Thrombocytopenia, AKI with vancomycin	Pseudomonas, anaerobes
Cefepime	2 g IV q8h	**Neurotoxicity/encephalopathy** (renal failure)	Pseudomonas • renal adjust
Meropenem	1 g IV q8h (2 g meningitis)	Seizures (less than imipenem)	**↓ valproate levels** • ESBL
Vancomycin	15–20 mg/kg q8–12h • load 25–30 mg/kg • **AUC 400–600**	**Nephrotoxicity**, vancomycin infusion reaction (slow infusion), ototoxicity	MRSA • AKI ↑ with pip-tazo, aminoglycosides
Linezolid	600 mg PO/IV q12h	**Thrombocytopenia** (> 2 weeks), **serotonin syndrome** (MAOI), peripheral/optic neuropathy, lactic acidosis, hypoglycemia	**Avoid SSRIs, tyramine-rich food** • 100% oral bioavailability
Amikacin / gentamicin / tobramycin	Amikacin 15–20 mg/kg • gent/tobra 5–7 mg/kg daily	**Nephrotoxicity**, **ototoxicity**, neuromuscular blockade	Level monitoring • avoid with loop diuretics, vancomycin
Oseltamivir	**75 mg BID × 5 d** • prophylaxis 75 mg daily • CrCl 30–60: 30 mg BID	N/V (with food), neuropsychiatric	Pregnancy-preferred • ↓ LAIV efficacy
Zanamivir	10 mg (2 inhalations) BID × 5 d	**Bronchospasm**	Avoid asthma/COPD
Baloxavir	40 mg (< 80 kg) or 80 mg once	Diarrhea	**Avoid dairy, Ca, antacids**
Paxlovid (nirmatrelvir/ritonavir)	**300/100 mg BID × 5 d** (eGFR 30–60: 150/100)	Dysgeusia, diarrhea	**Strong CYP3A4 inhibitor** → many DDIs

PART C — TUBERCULOSIS (TB)

8. TB Basics and Diagnosis

- **Mycobacterium tuberculosis** — acid-fast bacillus (AFB), slow growing, **airborne** (droplet nuclei from coughing) • ~1/4 of the world population infected.
- **Latent TB infection (LTBI):** infected, **asymptomatic**, **not contagious**, normal CXR, positive TST/IGRA • 5–10% lifetime risk of reactivation (higher in HIV ~10%/year).
- **Active TB disease:** symptomatic and **contagious** (pulmonary) • **risk factors for progression: HIV** (strongest), recent infection (< 2 years), **TNF-α inhibitors** (infliximab, adalimumab, etanercept), **corticosteroids ≥ 15 mg/day ≥ 1 month**, transplant, silicosis, diabetes, CKD/dialysis, malnutrition, smoking, alcohol, gastrectomy, young children.
- **Symptoms: cough > 2–3 weeks**, **hemoptysis**, **fever**, **night sweats**, **weight loss**, fatigue, chest pain • **CXR: upper lobe infiltrates with cavitation** (reactivation) • extrapulmonary: lymph nodes (scrofula), pleura, **meninges**, **spine (Pott's disease)**, miliary, GU, pericardium.

Tests for TB infection (TST and IGRA)

Induration (TST)	Positive for
≥ 5 mm	**HIV**, **recent close contact** of active TB, **fibrotic changes on CXR** (old TB), **organ transplant**, **immunosuppressed** (prednisone ≥ 15 mg/day ≥ 1 month, TNF-α inhibitors)
≥ 10 mm	**Recent immigrants (≤ 5 years) from high-prevalence countries**, **injection drug users**, residents/employees of **high-risk congregate settings** (prisons, nursing homes, homeless shelters, **healthcare facilities**), mycobacteriology lab staff, **children < 4 years** or exposed to high-risk adults, comorbidities (**diabetes**, CKD, silicosis, leukemia, gastrectomy, low body weight)
≥ 15 mm	**Persons with no risk factors**

- **TST (Mantoux): 0.1 mL (5 TU) PPD intradermal** (forearm) → **read in 48–72 h** → measure **induration, not erythema** • needs a return visit.
- **False positives: BCG vaccination**, nontuberculous mycobacteria • **false negatives:** immunosuppression (HIV, steroids), **recent live vaccine** (MMR, varicella — give TST the same day or wait 4 weeks), recent infection (< 8–10 weeks), malnutrition, overwhelming TB, technique.
- **Two-step TST** for baseline testing of adults who will be retested periodically (**healthcare workers**, nursing home residents) → detects **booster phenomenon**.
- **IGRA** (**QuantiFERON-TB Gold Plus**, T-SPOT.TB): blood test, single visit, **not affected by BCG** → **preferred in BCG-vaccinated persons** and those unlikely to return • acceptable/preferred for age ≥ 2 years (TST preferred in < 2 years in some guidance).
- **Neither TST nor IGRA distinguishes latent from active** → positive test → **CXR** and symptom evaluation to exclude active TB **before treating LTBI**.

Diagnosis of active TB

- **Sputum × 3** (8–24 h apart, ≥ 1 early morning) for **AFB smear** (rapid, low sensitivity) and **culture** (**gold standard**, 2–8 weeks; needed for susceptibility).
- **Nucleic acid amplification test (NAAT, Xpert MTB/RIF)** on at least one specimen → rapid diagnosis and detects **rifampin resistance (rpoB)**.
- **Infection control: airborne isolation** in a **negative-pressure room**, **N95 respirator** for staff, surgical mask for patient during transport • discontinue isolation when on effective therapy ≥ 2 weeks, clinically improving, and **3 negative AFB smears** (or TB ruled out).
- **Report** to public health; contact investigation • **directly observed therapy (DOT)** = standard of care for active TB.

9. Treatment of Latent TB Infection (CDC 2020)

Regimen	Dose	Notes
3HP (preferred): **isoniazid + rifapentine weekly × 12 doses** (3 months)	INH 15 mg/kg (max 900 mg) + rifapentine weight-based (max 900 mg) once weekly	Age ≥ 2 years • DOT or self-administered • flu-like/hypersensitivity reactions • **not in pregnancy** • many DDIs (inducer)
4R (preferred): **rifampin daily × 4 months**	**10 mg/kg (max 600 mg) daily** (children 15–20 mg/kg)	Less hepatotoxic than INH • higher completion • **DDIs** (potent inducer) • contacts of INH-resistant TB
3HR (preferred): **isoniazid + rifampin daily × 3 months**	INH 5 mg/kg (max 300) + RIF 10 mg/kg (max 600)	Also for children
6H or 9H (alternative): **isoniazid daily × 6–9 months**	**5 mg/kg (max 300 mg) daily** (or 900 mg twice weekly DOT)	When rifamycins not possible (DDIs — e.g., HIV ART, transplant) • **pregnancy** • 9H preferred in HIV/children/fibrotic lesions • + **pyridoxine**

- **Pyridoxine (vitamin B6) 25–50 mg daily** with isoniazid in: **pregnancy/breastfeeding**, **diabetes**, **HIV**, alcoholism, malnutrition, CKD, seizure disorder, peripheral neuropathy, elderly • prevents **INH peripheral neuropathy**.
- Monitor: baseline LFTs if risk (alcohol, liver disease, HIV, pregnancy/postpartum, hepatotoxic drugs) → monthly symptom review • stop if **ALT > 3× ULN with symptoms or > 5× ULN without**.
- Contacts of **MDR-TB**: fluoroquinolone (levofloxacin) × 6 months.

10. Treatment of Active Drug-Susceptible TB (ATS/CDC/IDSA 2016)

- **Always combination therapy** (prevents resistance) • **never add a single drug to a failing regimen** • **DOT** recommended • begin **RIPE** empirically while awaiting culture.
- **Standard 6-month regimen: Intensive phase 2 months: RIPE** (**R**ifampin + **I**soniazid + **P**yrazinamide + **E**thambutol) daily → **Continuation phase 4 months: isoniazid + rifampin** (total 6 months).
- **Ethambutol can be stopped** once isolate is shown **susceptible to INH and RIF**.
- **Extend continuation phase to 7 months (total 9 months)** if **cavitation on initial CXR AND positive sputum culture at 2 months** (also for HIV not on ART, and when PZA not used in intensive phase).
- **4-month regimen (CDC 2022)**: **rifapentine + moxifloxacin + isoniazid + pyrazinamide × 8 weeks → rifapentine + moxifloxacin + isoniazid × 9 weeks** (age ≥ 12, weight ≥ 40 kg, drug-susceptible pulmonary TB; not in pregnancy, HIV with CD4 < 100, extrapulmonary).
- **Monitoring: monthly sputum smear/culture** until 2 consecutive negative cultures • baseline **LFTs**, SCr, CBC, uric acid, HIV, hepatitis B/C, diabetes screen • **baseline and monthly visual acuity and red-green color vision** (ethambutol) • monthly clinical review for hepatotoxicity • weight-based dose adjustments.
- **Culture positive after 4 months = treatment failure** → suspect resistance/nonadherence → susceptibility testing, add ≥ 2 new drugs.

Drug	Daily dose (adults)	Weight bands (40–55 / 56–75 / 76–90 kg)
Isoniazid (INH)	**5 mg/kg (max 300 mg)**	300 mg
Rifampin (RIF)	**10 mg/kg (max 600 mg)**	600 mg
Pyrazinamide (PZA)	**20–25 mg/kg**	**1,000 / 1,500 / 2,000 mg**
Ethambutol (EMB)	**15–20 mg/kg**	**800 / 1,200 / 1,600 mg**
Rifabutin	5 mg/kg (300 mg)	Replaces RIF with protease inhibitors (HIV)
Rifapentine	1,200 mg daily (4-month regimen) • weekly in 3HP	—

TB drug toxicities (very high-yield)

Drug	Main ADRs	Interactions / notes
Isoniazid	**Hepatotoxicity** (↑ with age, alcohol, rifampin), **peripheral neuropathy** (B6 deficiency → **pyridoxine**), **drug-induced lupus**, rash, CNS (seizures in overdose → pyridoxine), **slow acetylators** ↑ toxicity	**CYP2C19/3A4 inhibitor** → ↑ **phenytoin**, carbamazepine, warfarin • take on empty stomach • avoid tyramine/histamine foods (mild MAO inhibition)
Rifampin	**Orange-red discoloration** of urine, tears, sweat, saliva (**stains soft contact lenses**), **hepatotoxicity** (cholestatic), **flu-like syndrome**, thrombocytopenia, hemolysis, interstitial nephritis, GI upset	**Potent inducer of CYP3A4, 2C9, 2C19, 1A2, P-gp, UGT** → ↓ levels of **warfarin**, **oral contraceptives** (use backup), **azoles**, **protease inhibitors**, **DOACs**, **methadone** (withdrawal), **tacrolimus/cyclosporine**, steroids, statins, sulfonylureas, levothyroxine, phenytoin • empty stomach
Pyrazinamide	**Hepatotoxicity** (most hepatotoxic of first-line), **hyperuricemia → gout**, arthralgia, rash, photosensitivity, GI	Avoid in acute gout • renal adjust (3×/week after dialysis)
Ethambutol	**Optic neuritis** → ↓ **visual acuity**, **red-green color blindness** (dose-related, reversible if early), peripheral neuropathy, hyperuricemia	**Baseline and monthly vision testing** • caution in young children who can't report vision • **renal adjust** (3×/week)
Rifabutin	**Uveitis**, neutropenia, orange secretions, arthralgia	Weaker inducer than rifampin (used with PIs) • ↑ levels with PIs (↓ rifabutin dose)
Rifapentine	Same as rifampin, hypersensitivity/flu-like (3HP)	Potent inducer
Streptomycin / amikacin	**Ototoxicity** (vestibular — streptomycin), **nephrotoxicity**	**Avoid in pregnancy** (fetal deafness)
Fluoroquinolones (moxifloxacin, levofloxacin)	Tendinopathy, QT, CNS, dysglycemia	MDR-TB, 4-month regimen, hepatotoxicity alternative
Bedaquiline	**QT prolongation** (boxed), hepatotoxicity, ↑ mortality (boxed, early data)	MDR-TB • CYP3A4 substrate (avoid rifampin)
Linezolid (MDR)	**Myelosuppression**, **peripheral and optic neuropathy**, lactic acidosis, serotonin syndrome	MDR-TB (BPaLM)
Pretomanid	Hepatotoxicity, myelosuppression, neuropathy (with linezolid)	Only in BPaL(M) regimen

Drug-induced hepatitis during TB treatment: stop all hepatotoxic drugs if **ALT ≥ 3× ULN with symptoms** (nausea, jaundice, abdominal pain) or **≥ 5× ULN without symptoms** or bilirubin elevated • give non-hepatotoxic drugs if TB severe (**ethambutol + FQ + aminoglycoside/linezolid**) • when ALT < 2× ULN → **reintroduce sequentially**: **rifampin (± ethambutol) first** → then **isoniazid** after 3–7 days → **PZA last** (or omit and extend to 9 months).

Special populations

- **Pregnancy: INH + RIF + EMB** × 2 months, then INH + RIF × 7 months (total 9 months) — **PZA not routinely used in US** (WHO allows) • + pyridoxine • **avoid streptomycin/aminoglycosides** (ototoxicity), FQs, **rifapentine** • breastfeeding OK (+ B6 for infant if on INH).
- **HIV:** same 6-month regimen (daily) • start **ART within 2 weeks if CD4 < 50**, within 8 weeks otherwise • **TB meningitis: delay ART 4–8 weeks** (↑ mortality with early ART) • **rifampin interacts with PIs and many INSTIs** → use **rifabutin with PIs**, **dolutegravir 50 mg BID** with rifampin, efavirenz OK • watch for **IRIS** (paradoxical worsening → steroids).
- **Renal failure / dialysis:** INH and RIF normal doses • **PZA 25–35 mg/kg and EMB 15–25 mg/kg three times weekly after dialysis**.
- **Liver disease:** fewer hepatotoxic drugs (e.g., RIF + EMB + FQ, avoid PZA) • close LFT monitoring.
- **Children:** higher mg/kg doses (INH 10–15, RIF 10–20, PZA 30–40, EMB 15–25 mg/kg) • ethambutol acceptable with monitoring.

- **Extrapulmonary:** mostly **6 months • TB meningitis 9–12 months + adjunctive dexamethasone** (6–8 weeks taper) • bone/joint (Pott's) 6–9 months • **pericarditis: steroids not routine** (selected).
- **Drug-resistant TB: MDR-TB** = resistant to **INH and RIF • pre-XDR** (+ FQ) • **XDR** (+ FQ + bedaquiline/linezolid) • WHO preferred: **BPaLM** (**b**edaquiline + **p**retomanid + **l**inezolid + **m**oxifloxacin) × **6 months** • INH mono-resistance → RIF + EMB + PZA + levofloxacin × 6 months.
- **BCG vaccine**: live attenuated, given at birth in high-prevalence countries (e.g., Saudi Arabia) → protects children against severe forms (TB meningitis, miliary) • **contraindicated in immunocompromised/HIV**, pregnancy.

11. TB Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Isoniazid	**5 mg/kg (300 mg) daily** • 3HP 15 mg/kg weekly (max 900)	**Hepatitis**, **peripheral neuropathy**, lupus	**Pyridoxine 25–50 mg** • ↑ phenytoin • empty stomach
Rifampin	**10 mg/kg (600 mg) daily** • 4R LTBI	**Orange body fluids**, hepatitis, flu-like, thrombocytopenia	**Potent inducer** → ↓ warfarin, OCs, azoles, PIs, DOACs, methadone, tacrolimus
Pyrazinamide	**20–25 mg/kg** (1–2 g) daily × 2 months	**Hepatitis** (most), **hyperuricemia/gout**, arthralgia	Avoid in pregnancy (US) • renal 3×/week
Ethambutol	**15–20 mg/kg** (0.8–1.6 g) daily	**Optic neuritis** (acuity, red-green)	Monthly vision • renal adjust • stop when INH/RIF susceptible
Rifapentine	3HP weekly (max 900 mg) • 1,200 mg daily (4-month)	Flu-like, hepatitis, orange fluids	Inducer • not in pregnancy
Rifabutin	300 mg daily (150 mg with boosted PIs)	**Uveitis**, neutropenia	Use with PIs (HIV)
Moxifloxacin / levofloxacin	400 mg / 750–1,000 mg daily	QT, tendons	4-month regimen, MDR-TB, MDR contacts
Bedaquiline	400 mg daily × 2 weeks → 200 mg 3×/week (or 200 mg daily × 8 wk → 100 mg)	**QT**, hepatotoxicity	MDR-TB • with food
Pretomanid	200 mg daily	Hepatotoxicity	BPaL(M)
Pyridoxine	25–50 mg daily (100 mg if neuropathy)	Neuropathy at very high doses	With INH

PART D — BRUCELLOSIS

12. Brucellosis (Malta fever, undulant fever)

- **Zoonosis** caused by **Brucella** — small **gram-negative coccobacilli**, **facultative intracellular** (survive in macrophages → need intracellular-penetrating drugs and long combination therapy) • **endemic in the Middle East (Saudi Arabia)**, Mediterranean, Central Asia, Latin America.
- **Species: B. melitensis** (goats, sheep, camels — **most common and most virulent**), B. abortus (cattle), B. suis (pigs), B. canis (dogs).
- **Transmission: unpasteurized (raw) milk and dairy products** (fresh cheese, camel milk — most common), undercooked meat/liver, direct contact with infected animals (**abortion/birth products** — farmers, veterinarians, butchers), **aerosol** (**laboratory workers** — one of the most common lab-acquired infections), rarely person-to-person (breastfeeding, sexual, transplant).
- **Incubation:** 1–4 weeks (up to months).

Clinical features

- **Fever** (often **undulant**/intermittent), **profuse malodorous sweats** (especially at night), **fatigue/malaise**, **arthralgia**, myalgia, **back pain**, headache, anorexia, weight loss, depression.

- Signs: **hepatomegaly**, **splenomegaly**, **lymphadenopathy**.
- **Labs:** mild ↑ LFTs, **anemia, leukopenia (relative lymphocytosis)**, thrombocytopenia (pancytopenia), normal/low WBC.
- **Classified as:** acute (< 2 months), subacute (2–12 months), chronic (> 1 year).

Focal complications

Complication	Notes
Osteoarticular (most common, 20–60%)	**Sacroiliitis** (young adults), **spondylitis** (lumbar spine, older adults — can cause paravertebral/epidural abscess), peripheral arthritis (knee, hip), osteomyelitis
Genitourinary	**Epididymo-orchitis** (men), pyelonephritis, abortion in pregnancy (miscarriage, preterm delivery)
Neurobrucellosis (< 5%)	Meningitis, meningoencephalitis, cranial nerve palsies (hearing loss), myelitis, brain abscess
Endocarditis (~1–2%)	**Main cause of death** (aortic valve) • usually needs **valve surgery** + prolonged antibiotics
Others	Hepatitis/hepatic abscess, pneumonia, uveitis, skin lesions, cytopenias

Diagnosis

- **Blood culture** = gold standard (positive in 40–70% acute; bone marrow culture more sensitive) • slow growing → **inform the lab** (biosafety risk, prolonged incubation; automated systems usually positive in 3–7 days).
- **Serology: serum (standard) agglutination test (SAT)** — titer **≥ 1:160** (≥ 1:320 in endemic areas) or **4-fold rise** • **Rose Bengal** (rapid screening) • **2-mercaptoethanol (2-ME) test** (IgG — active/chronic infection) • **Coombs (antibrucella) test** (incomplete antibodies — chronic/relapse) • **ELISA** (IgM, IgG) • **PCR** (rapid, sensitive).
- B. canis is not detected by standard SAT.
- **Imaging** (MRI) for spondylitis • echocardiography if murmur • LP if neurologic signs.

Treatment (WHO / international consensus)

Principles: combination therapy (monotherapy → high relapse) with drugs that achieve **intracellular** concentrations • **prolonged duration (≥ 6 weeks)** • **doxycycline is the backbone**.

Form	Recommended regimen
Uncomplicated (adults, non-pregnant)	**Doxycycline 100 mg PO BID × 6 weeks + streptomycin 15 mg/kg (1 g) IM daily × 2–3 weeks** (most effective, lowest relapse) **OR doxycycline 100 mg BID × 6 weeks + gentamicin 5 mg/kg IV/IM daily × 7 days** (aminoglycoside alternative) **OR doxycycline 100 mg BID + rifampin 600–900 mg (15 mg/kg) PO daily, both × 6 weeks** (all-oral, WHO; slightly higher relapse)
Spondylitis / osteoarticular	**Doxycycline + rifampin for ≥ 12 weeks (3 months) + streptomycin 2–3 weeks or gentamicin 1–2 weeks** (triple therapy) • longer (up to 6 months) for complicated spondylitis/abscess • surgical drainage if needed
Neurobrucellosis	**Ceftriaxone 2 g IV q12h** (first 4–6 weeks) **+ doxycycline + rifampin** (± TMP-SMX) for **3–6 months** (until CSF normalizes)
Endocarditis	**Aminoglycoside (gentamicin) first 2–4 weeks + doxycycline + rifampin** (± TMP-SMX or ceftriaxone) for **≥ 6 weeks to 3–6 months** + **early valve replacement** in most patients
Children < 8 years	**TMP-SMX (TMP 5 mg/kg BID) + rifampin (15–20 mg/kg/day) × 6 weeks** (± gentamicin 5 mg/kg × 7 days) • avoid doxycycline (teeth) • ≥ 8 years: as adults
Pregnancy	**Rifampin 900 mg daily × 6 weeks** (± **TMP-SMX** — avoid in 1st trimester and **after 36 weeks**/near term) • **avoid doxycycline** (teeth/bone) and **streptomycin** (fetal ototoxicity) • ceftriaxone may be added
Relapse (5–15%, usually within 6 months)	Usually due to inadequate duration/adherence (not resistance) → **repeat the same regimen** (or a more effective one) • look for focal disease
Post-exposure prophylaxis (lab exposure)	**Doxycycline 100 mg BID + rifampin 600 mg daily × 3 weeks** (TMP-SMX + rifampin if pregnant/doxy CI) + serologic monitoring

- **Rifampin** ↓ doxycycline levels (induction) — part of why doxy + streptomycin is more effective • rifampin ↓ oral contraceptives and warfarin effect • orange fluids.
- **Doxycycline:** take with full glass of water, stay upright (esophagitis), photosensitivity, separate from Ca/Fe/Mg/antacids/dairy.
- **Streptomycin/gentamicin:** monitor renal function, hearing/vestibular function, levels.
- **Prevention: pasteurize/boil milk**, avoid raw dairy and undercooked meat, gloves/PPE for animal handlers and labs, **animal vaccination** (no human vaccine) • **notifiable disease**.

13. Brucellosis Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Doxycycline	**100 mg PO BID × 6 weeks** (≥ 12 weeks spondylitis)	Esophagitis, **photosensitivity**, GI	Backbone • avoid < 8 years and pregnancy • chelation • rifampin ↓ levels
Rifampin	**600–900 mg (15 mg/kg) daily × 6 weeks**	Orange fluids, hepatitis, flu-like	**Inducer** (OCs, warfarin) • pregnancy option
Streptomycin	**15 mg/kg (1 g) IM daily × 2–3 weeks**	**Ototoxicity (vestibular)**, nephrotoxicity	Most effective combination with doxy • avoid pregnancy
Gentamicin	**5 mg/kg IV/IM daily × 7 days** (endocarditis longer)	Nephro/ototoxicity	Alternative to streptomycin • levels
TMP-SMX	TMP 5 mg/kg BID (adult DS BID)	Rash, hyperkalemia, marrow	Children < 8, pregnancy (not 1st trimester/near term)
Ceftriaxone	2 g IV q12h	Biliary sludge	Neurobrucellosis (CNS penetration)
Ciprofloxacin	500 mg BID	FQ class effects	Second-line/alternative partner (higher relapse)

Exam Traps and Frequently Tested Points

1. **E. coli** causes most UTIs • **S. saprophyticus** in young women • **Proteus** → struvite stones and **nitrofurantoin-resistant** • nitrite negative with Enterococcus/S. saprophyticus.
2. **Uncomplicated cystitis: nitrofurantoin 100 mg BID × 5 days**, **TMP-SMX DS BID × 3 days** (resistance < 20%), **fosfomycin 3 g once** • FQs reserved.
3. **Nitrofurantoin and fosfomycin are NOT for pyelonephritis** • nitrofurantoin avoid CrCl < 30 • pulmonary, hepatic, neuropathy toxicity.
4. **Pyelonephritis:** cipro 500 BID × 7 d or levo 750 × 5–7 d (if FQ resistance ≤ 10%, else initial ceftriaxone) • TMP-SMX 14 d if susceptible • ESBL → carbapenem.
5. **ASB:** treat only in **pregnancy** and before **urologic procedures with mucosal bleeding** • NOT in elderly, diabetics, catheters.
6. **Pregnancy UTI:** nitrofurantoin, cephalexin, amoxicillin-clavulanate, fosfomycin • **avoid FQs, tetracyclines, TMP-SMX (1st trimester/term)** • pyelo → admit, IV ceftriaxone.
7. **Men:** 7 days • prostatitis: FQ or TMP-SMX 2–4 weeks (chronic 4–6 weeks FQ) • CAUTI: change catheter, 7 days.
8. **Phenazopyridine** ≤ 2 days, orange urine, G6PD hemolysis • **methenamine** needs acidic urine.
9. **TMP-SMX:** hyperkalemia, ↑ INR, ↑ SCr (secretion), SJS • **FQ:** boxed tendon/neuropathy/CNS/MG, QT, chelation, ciprofloxacin–tizanidine CI.
10. **GAS pharyngitis:** penicillin V or amoxicillin × **10 days** (benzathine 1.2 MU once) • allergy: cephalexin (mild) / clindamycin or azithromycin 5 d (severe) • **negative RADT in children → throat culture**.
11. **AOM: high-dose amoxicillin 80–90 mg/kg/day** • amox-clav if amoxicillin in 30 days or conjunctivitis • duration 10 d (< 2 y), 7 d (2–5 y), 5–7 d (≥ 6 y) • observation option.
12. **Sinusitis bacterial if:** ≥ 10 days, severe ≥ 3–4 days (fever ≥ 39 + purulence), double sickening • **amox-clav first-line** 5–7 days adults • **no macrolides/TMP-SMX**.
13. **Common cold and acute bronchitis → no antibiotics** • pertussis → azithromycin.

14. **Influenza:** oseltamivir 75 mg BID × 5 d (≤ 48 h, but any time if hospitalized/high risk) • CrCl 30–60: 30 mg BID • zanamivir → bronchospasm • baloxavir single dose, no dairy.
15. **CAP outpatient healthy:** amoxicillin 1 g TID, doxycycline, or macrolide (if resistance < 25%) • **comorbidities:** amox-clav/cephalosporin + macrolide/doxy, or resp FQ.
16. **CAP inpatient:** β-lactam + macrolide or resp FQ alone • **ICU:** β-lactam + macrolide or β-lactam + FQ • add MRSA/Pseudomonas only with specific risk factors.
17. **CAP duration ≥ 5 days** (stable 48–72 h) • no routine anaerobic coverage for aspiration • procalcitonin not to withhold antibiotics • **CURB-65**: 0–1 home, 2 admit, ≥ 3 ICU.
18. **β-lactams don't cover atypicals** (Mycoplasma, Chlamydophila, Legionella) → macrolide, doxycycline, FQ • Legionella → hyponatremia, diarrhea.
19. **HAP/VAP:** ≥ 48 h after admission/intubation • cover S. aureus + Pseudomonas • MRSA (vanco/linezolid) and double antipseudomonal only with risk factors • **7 days**.
20. **Daptomycin never for pneumonia** (surfactant) • **moxifloxacin not for UTI** • **linezolid**: thrombocytopenia, serotonin syndrome, tyramine.
21. **TST cut-offs: ≥ 5 mm** HIV, contacts, transplant, immunosuppressed • **≥ 10 mm** immigrants, IDU, HCW, < 4 years, diabetes • **≥ 15 mm** no risk.
22. **TST read 48–72 h (induration)** • **IGRA preferred if BCG-vaccinated** • two-step TST for HCW baseline • positive test → **CXR** to rule out active TB.
23. **LTBI: 3HP** (INH + rifapentine weekly × 12), **4R**, **3HR** (preferred) • 6–9H alternative • **pyridoxine** with INH.
24. **Active TB: RIPE × 2 months → INH + RIF × 4 months** (6 months total) • **9 months** if cavitation + positive culture at 2 months • stop EMB when susceptible • DOT.
25. **Sputum × 3** AFB smear + culture (gold standard) + NAAT • airborne isolation, negative pressure, N95 • 3 negative smears to discontinue.
26. **INH:** hepatitis, **neuropathy (B6)**, lupus, ↑ phenytoin • **RIF: orange fluids (contact lenses)**, potent inducer (OCs, warfarin, PIs, azoles, DOACs, methadone).
27. **PZA:** hepatotoxicity (most), **hyperuricemia/gout** • **EMB: optic neuritis** (visual acuity, **red-green**) → monthly vision checks • PZA/EMB 3×/week in dialysis.
28. **Hepatotoxicity:** stop if ALT > 3× with symptoms or > 5× without → reintroduce RIF → INH → PZA last.
29. **TB in pregnancy:** INH + RIF + EMB × 9 months (no PZA in US) • avoid streptomycin, rifapentine • **HIV:** rifabutin with PIs, dolutegravir 50 mg BID with rifampin, ART within 2 weeks if CD4 < 50 (delay in TB meningitis).
30. **TB meningitis:** 9–12 months + dexamethasone • **MDR-TB** (INH + RIF resistant) → **BPaLM × 6 months** • bedaquiline QT.
31. **Brucellosis:** B. melitensis (goats/sheep/camels) • **raw milk** • undulant fever, **malodorous sweats**, arthralgia, hepatosplenomegaly • **sacroiliitis/spondylitis** most common complication • endocarditis = main cause of death.
32. **Brucellosis diagnosis:** blood culture (alert lab — lab-acquired infection) • SAT ≥ 1:160 (≥ 1:320 endemic) • 2-ME, Coombs • PCR.
33. **Brucellosis treatment: doxycycline × 6 weeks + streptomycin 2–3 weeks** (best) or **+ gentamicin 7 days** or **+ rifampin × 6 weeks** • **never monotherapy**.
34. **Spondylitis ≥ 12 weeks** (doxy + rif + aminoglycoside) • **neurobrucellosis:** ceftriaxone + doxy + rif 3–6 months • **endocarditis:** aminoglycoside + doxy + rif + surgery.
35. **Brucellosis in children < 8:** TMP-SMX + rifampin • **pregnancy:** rifampin ± TMP-SMX (avoid doxy, streptomycin) • relapse → repeat regimen • lab PEP: doxy + rif × 3 weeks.

Note: doses and recommendations follow current IDSA, ATS/IDSA, CDC, AAP, and WHO guidance. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.